

Killer Sudoku — Project Documentation

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1. Mockup overview & design rationale

The app ships with eight distinct pages. The visual language is intentionally modern but restrained: dark theme, animated aurora background, glassmorphism on cards, electric violet → cyan accent gradient, generous whitespace, and crisp monospace numerals inside the grid.

Page-by-page mockup notes

Page	Path	Key elements
Landing	/	Hero with a 4x4 mini-grid preview, animated headline, three stat cards (1 / 405 / 3), feature cards below
Sign in	/auth/login	Single-card glass form, accent-gradient submit
Register	/auth/register	3-field form with inline rules ("3–20 chars / letter+digit / valid email")
Puzzle list	/play	Difficulty filter pills, card grid with badge + creator + plays + average rating
Solve	/play/[id]	Grid + sidebar (NumberPad, Hint button, Check, Restart). Timer + hint counter in header. Victory card with rating UI on win.
Builder	/create	Same 9x9 grid, sidebar lists current cage + cage list + difficulty selector + Save. Cage cells colour-coded.
Daily	/daily	Single highlighted puzzle + daily leaderboard table
Leaderboard	/leaderboard	Global leaderboard table with difficulty filter
Stats	/stats	4 KPI cards + Recharts bar chart (best vs avg per difficulty) + recent-results table
Rules	/rules	Five rule cards + a "useful fact" card explaining the $\Sigma=405$ invariant

Design system at a glance

Element	Value
Background	#07060d (near-black) + animated aurora blobs
Accent A	#7c3aed (violet)
Accent B	#06b6d4 (cyan)
Accent gradient	linear-gradient(135deg, accentA → accentB)
Glass card	rgba(255,255,255,0.04) + 1 px border + 12 px backdrop blur

Element	Value
Mono font	ui-monospace, SFMono-Regular, Menlo
Animation	Framer Motion for entry/transition, CSS keyframes for cell pop / shake / sparkle
Sound	Web-Audio-generated tones (no external assets)
Icons	lucide-react

Cell colours inside the grid are derived from the cage id using a fixed 8-stop HSL palette and mixed at 8 % opacity with the background so that adjacent cages stay readable.

2. Database diagram (ER)

```

erDiagram
    USERS ||--o{ PUZZLES : creates
    USERS ||--o{ RESULTS : "plays / submits"
    USERS ||--o{ RATINGS : rates
    PUZZLES ||--o{ RESULTS : has
    PUZZLES ||--o{ RATINGS : receives
    PUZZLES ||--o{ DAILY_PUZZLES : "featured on"

    USERS {
        int id PK
        varchar username UK
        varchar email UK
        varchar password_hash
        timestamp created_at
    }
    PUZZLES {
        int id PK
        int creator_id FK
        tinyint difficulty
        json grid_json
        json cages_json
        timestamp created_at
    }
    RESULTS {
        int id PK
        int user_id FK
        int puzzle_id FK
        int time_seconds
        int hints_used
        timestamp completed_at
    }
    RATINGS {
        int id PK
        int user_id FK
        int puzzle_id FK

```

```

    tinyint stars
    enum difficulty_feedback
    timestamp created_at
  }
  DAILY_PUZZLES {
    int id PK
    int puzzle_id FK
    date date UK
  }

```

The Prisma schema (`prisma/schema.prisma`) is the source of truth. The raw MySQL DDL is shipped as `sudoku.sql` and matches the Prisma schema 1:1 (incl. the unique `(user_id, puzzle_id)` index for one rating per user per puzzle).

3. Class diagram (core)

```

classDiagram
    class Cage {
        +int id
        +int sum
        +Array~[number,number]~ cells
    }
    class Puzzle {
        +Grid grid
        +Cage[] cages
        +1|2|3 difficulty
    }
    class Solver {
        +solve(grid, cages) Grid?
        +countSolutions(grid, cages, max) int
        -buildWorkspace()
        -isValid(r, c, v) bool
        -cageAllows(r, c, v) bool
        -findMRV() Cell?
    }
    class Validator {
        +registerSchema
        +loginSchema
        +resultSchema
        +ratingSchema
        +puzzleCreateSchema
        +validateCageSumTotal(cages) bool
        +validateCageStructure(cages, fullCover) StructureResult
        +checkSolution(grid, cages) CheckResult
    }
    class Hint {
        +computeHint(grid, cages) Hint?
    }

```

```

class Scoring {
  +scoreOf(result) int
  +pickDailyPuzzleId(date, pool) int
  +computeStats(results) Stats
  +computeStreak(days, today) int
}
class GameStore {
  +Grid grid
  +Set~int~[][] notes
  +Cage[] cages
  +int hintsUsed
  +loadPuzzle()
  +placeNumber(n)
  +applyHint(r, c, v)
  +tick()
}
Hint --> Solver : calls solve()
Solver ..> Cage
Validator ..> Cage
GameStore ..> Cage
Puzzle --> Cage

```

4. Three additional use cases (justification)

#	UC	Why it's worth adding
12	Puzzle of the Day	Daily competitive challenge — every user gets the same puzzle picked deterministically from a stable hash of the date. Drives repeat engagement and a daily leaderboard.
13	Rate puzzle after solving	Lets the community surface the best puzzles and provides puzzle creators with feedback (stars + qualitative "too easy / fits / too hard").
14	Personal statistics dashboard	Long-term motivation: best time per difficulty, current daily streak, total hints used, recent-solve history. Visualised with Recharts.

All three are fully implemented and tested.

5. Validation rules (overview)

Field / object	Rule	Source
username	3–20 chars, [A–Za–z0–9_] only, unique	registerSchema + DB unique key
email	RFC-ish email format, unique	registerSchema + DB unique key
password	≥ 8 chars, ≥ 1 letter, ≥ 1 digit, bcrypt-hashed	registerSchema , hashPassword
difficulty	exactly 1, 2 or 3	Zod union of literals + DB CHECK

Field / object	Rule	Source
<code>cage.sum</code>	$\text{int } 1 \leq \text{sum} \leq 45$	Zod <code>min(1).max(45)</code>
<code>cage.cells</code>	1–9 elements; all within 0–8 row/col; orthogonally connected	Zod + builder UI
Cage structure	No two cages may overlap	<code>validateCageStructure</code>
Cage coverage	All 81 cells must belong to exactly one cage	<code>validateCageStructure(..., requireFullCover=true)</code>
Σ cage sums	Must equal 405	<code>validateCageSumTotal</code>
Puzzle uniqueness	Exactly one solution	<code>countSolutions(grid, cages, 2) === 1</code>
<code>stars</code>	$\text{int } 1\text{--}5$	Zod <code>min(1).max(5)</code> + DB CHECK
<code>difficulty_feedback</code>	one of <code>too_easy</code> / <code>fits</code> / <code>too_hard</code>	Zod enum + DB ENUM
<code>timeSeconds</code>	$\text{int } 0 \leq t \leq 86400$	Zod nonnegative + <code>max(86400)</code>
<code>hintsUsed</code>	$\text{int } 0 \leq h \leq 81$	Zod <code>min(0).max(81)</code>
One rating per user per puzzle	Yes — upsert	DB UNIQUE (user_id, puzzle_id)

6. Hint algorithm

```
function computeHint(grid, cages):
  if grid has no empty cell: return null
  solved ← solve(grid, cages)           // full backtracking solution
  if solved is null: return null

  bestCell ← null
  bestCount ← ∞

  for each empty cell (r, c):
    used ← { values in row r } ∪ { values in column c }
           ∪ { values in 3×3 box of (r,c) }
           ∪ { values in the cage containing (r,c) }
    candidates ← 9 − |used \ {0}|

    if candidates < bestCount:
      bestCount ← candidates
      bestCell ← (r, c)
      if bestCount = 1: return (r, c, solved[r][c]) // hard force, return immediate

  return (bestCell.r, bestCell.c, solved[bestCell.r][bestCell.c])
```

In other words: solve the full puzzle once with the backtracking solver, then return the value of the **most-constrained empty cell** (MRV — minimum remaining values). Returning an MRV cell makes hints feel "useful" because the solver picks the one a human would have solved next.

7. The $\Sigma = 405$ sanity check

Each row of a solved 9×9 Sudoku contains the digits 1...9 exactly once, so each row sums to $1+2+\dots+9 = 45$. The grid has 9 rows, so the total of all 81 cells is $9 \times 45 = 405$.

Cages partition the grid: every cell belongs to exactly one cage. Therefore **the sum of every cage's sum must also equal 405**. Any input where the cage sums total something other than 405 cannot possibly correspond to a valid solved grid — and we can reject it in $O(n)$ time without invoking the much more expensive solver.

The application uses this in three places:

1. `puzzleCreateSchema` validation in `/api/puzzles` POST — rejects with 400.
2. The puzzle builder's coverage panel — live counter.
3. The generator (`generatePuzzle`) — short-circuits a generation attempt before running `countSolutions`, dramatically improving generation speed.

8. Test protocol

The full protocol with 70 test cases (33 positive · 24 negative · 13 boundary) is shipped as `docs/test-protocol-FINAL.pdf`. 40 of the cases are automated Vitest tests in `tests/`. The remaining 30 are manual cases executed against the running app.

Final tally: 70 / 70 cases pass. See `docs/test-protocol-FINAL.pdf` for the per-row breakdown.

9. Beyond requirements — random puzzle generator

The `/create` page ships with two modes:

- **Build manually** — tap cells to draw cages, set sums, save. Cage sums must total 405 and the solver must confirm the puzzle has exactly one solution.
- **Generate random** — pick a difficulty, click *Generate*, then review and Save. The generator (`src/lib/generator.ts`) builds a solved grid via base-pattern + permutations, partitions the cells into adjacent cages weighted by the chosen difficulty, computes sums from the solved grid, and runs `countSolutions(..., max=2)` to reject ambiguous puzzles. Retries up to 200 attempts per call (typically < 50 ms for easy, up to ~3 s for hard).

The generator's API entry point is `POST /api/puzzles/generate` (auth required).

10. R1 fix pass — what changed since the initial submission

Bug / improvement	Fix
Hydration error on landing	<code><AuroraBackground></code> is now mounted client-only via <code>useEffect</code> + <code>mounted</code> state, and <code><html></code> / <code><body></code> carry <code>suppressHydrationWarning</code>
Navbar not updating after login	<code>Navbar</code> reads <code>['me']</code> from TanStack Query with <code>initialUser</code> as seed; login & register pages call <code>setQueryData(['me'], ...)</code> + <code>invalidateQueries(['me'])</code>
Cell box growing with content	Added <code>grid-template-rows: repeat(9, minmax(0,1fr))</code> , <code>box-sizing: border-box</code> , <code>min-width/height: 0</code> , <code>overflow: hidden</code> to <code>.sudoku-cell</code>
Hint +60s penalty	<code>gameStore.applyHint</code> recomputes <code>displaySeconds = elapsedSeconds + hintsUsed * 60</code> and bumps <code>hintPenaltyFlash</code> ; <code>Timer</code> shows a "+60s" framer-motion overlay
Hint thinks grid is complete with wrong values	<code>computeHint(current, cages, original)</code> solves the <i>original</i> grid (givens only), then picks any non-given cell where <code>current ≠ solution</code>
Card titles in mono	Switched titles to Geist Sans, semibold, tight tracking; only the small <code>#N</code> index stays mono
Sort order reversed	API now sorts <code>[{difficulty: 'asc'}, {createdAt: 'asc'}]</code>
Boring landing for guests	Replaced <code>/</code> with a 6-section <code><GuestLanding></code> (hero, features, how-it-works, stats strip with counter animations, final CTA, footer). Logged-in users redirect to <code>/play</code> .
Welcome banner & navbar	New gradient-initials avatar + dropdown menu (Stats / Create / Browse / Sign out); toast moved to bottom-right
Random puzzle generator	New mode toggle in <code>/create</code> + <code>POST /api/puzzles/generate</code> route

11. R2 pass — visual overhaul + give up + share + avatar upload

Area	Change
Color palette	New zinc/violet system: canvas <code>#0a0a0b</code> , card <code>#1a1a1f</code> , ink <code>#f4f4f5/#a1a1aa/#52525b</code> , single accent <code>#a78bfa</code> . Removed all violet→cyan gradients; logo uses solid colours + 3×3 dot-grid monogram (<code><LogoMark/></code>) instead of the sparkle icon.
Spacing	All page containers use <code>max-w-[1200px]</code> <code>mx-auto</code> <code>px-6</code> with <code>pt-16</code> so content sits clear of the navbar. Auth pages are vertically centred.
Bug — profile dropdown	Solid <code>#1a1a1f</code> background, 1 px white-10 border, scale-in entrance, z-50, no backdrop-blur.
Bug — hint cap	Frontend disables the Hint button when <code>hintsUsed ≥ emptyCellCount</code> (per-puzzle). Backend rejects with 400 + "Hint limit reached" if the client sends a <code>hintsUsed ≥</code> the empty-cell count of the original puzzle.
Bug — Daily hints	<code>/daily</code> reuses <code><SolveBoard></code> with <code>isDaily</code> flag → identical hint endpoint and code path as <code>/play/[id]</code> .

Area	Change
Bug — Daily leaderboard	After solve, <code>queryClient.invalidateQueries(['daily-lb'])</code> so the table updates instantly.
Bug — Grid borders	Reworked to a CSS-grid with <code>gap: 1 px</code> against a coloured backdrop — eliminates doubled lines. 3x3 box dividers via inset box-shadows; corners square.
Bug — Rating persistence	<code>prisma.rating.upsert</code> with the compound unique key now actually persists; the submit button cycles idle → submitting → saved (green check), stars turn <code>#fcd34d</code> after save.
Bug — Duplicate difficulty selector	Random-mode shows only the upper selector; manual mode only the lower. Same state behind both.
Polish — 3D landing grid	Hero grid tilts toward the cursor via framer-motion spring (<code>rotateX / rotateY</code>); returns to idle float on mouse leave.
Stats — editorial redesign	Large mono numerals (no cards), Line chart for "Average time by difficulty" (mint/amber/rose), Bar chart for "Puzzles per week", borderless table for recent solves with hairline separators. Backend now returns an 8-week series in <code>/api/stats</code> .
Feature — Give Up	Confirmation dialog → <code>GET /api/puzzles/[id]/solution</code> → <code>revealSolution(solved)</code> in the game store fills empty/wrong cells, marks them in rose, sets status <code>gave_up</code> . No result is saved.
Feature — Share	Modal with "Copy link" and "Copy as text" (clipboard API).
Feature — Avatar upload	<code>POST /api/auth/avatar</code> (multipart, ≤ 2 MB, jpeg/png/webp). <code>sharp</code> resizes to 256 × 256 webp at quality 85 and writes to <code>public/avatars/{userId}_{timestamp}.webp</code> . Old file is unlinked. <code>DELETE /api/auth/avatar</code> removes file + clears column. New <code>avatar_url VARCHAR(500)</code> column on <code>users</code> (Prisma + raw <code>sudoku.sql</code> both updated). <code><Avatar/></code> component is the single source of truth — used in Navbar dropdown (with hover-to-upload).

12. R3 pass — multi-accent palette, prefills, 3D landing, My Puzzles

Area	Change
Palette	Multi-accent: violet <code>#a78bfa</code> (primary), cyan <code>#22d3ee</code> (info / "You" / corner dots in the logo), amber <code>#fbbf24</code> (Daily / 405 highlight). Canvas warmed to <code>#0a0a0f</code> . 80/15/3/2 disciplined usage.
Logo	Two-tone 3x3 dot grid: four corner dots cyan, five cross-and-center dots violet.
Cage palette	Expanded from 8 to 12 colours (rose, orange, amber, lime, mint, cyan, sky, indigo, lavender, pink, peach, teal).
Grid borders	Cell borders are now clearly visible: 1 px gaps over a <code>rgba(255,255,255,0.14)</code> backdrop, 2 px outer border. 3x3 box dividers via <code>box-divider-right/-bottom</code> pseudo-element ribbons in white-34. Dashed cage edges render only where the cage actually ends. Cage-sum label uses the cage's full-opacity colour.

Area	Change
Pre-filled clues by difficulty	Easy: 20–25 given clues. Medium: 8–12. Hard: 0. Both the random generator and the manual <code>/api/puzzles</code> POST add prefills if the saved grid is empty. Clues are spread across all 9 3×3 boxes via <code>pickDistributedCells</code> . Pre-filled cells are immutable in the solve interface — keyboard, click-to-erase and the Erase button all ignore them.
Landing 3D grid	Multi-layer Framer-Motion build with floor shadow, base plate, raised filled cells (translateZ 6 px), cursor-tracked specular highlight, idle floating animation, prefers-reduced-motion fallback.
Rules page	Now uses <code>pt-24</code> and the editorial caption/H1 pattern.
Puzzle list redesign	Editorial layout: "Library" caption + 5xl H1, a highlighted Daily Challenge section, then a "Browse" grid of cards. Each card has a <code><MiniGridPreview/></code> thumbnail of its cage layout, a difficulty badge, play count, average rating with gold star, and a difficulty-coloured hover glow. The card author's name links them to "You" in cyan when the viewer is the creator.
My puzzles	New <code>/my-puzzles</code> route + dropdown link. <code>GET /api/puzzles/mine</code> returns the user's puzzles with aggregate play count + average rating. The page renders an editorial table with hover-revealed delete buttons.
Delete a puzzle	<code>DELETE /api/puzzles/[id]</code> — creator OR admin only. Cascade-delete results and ratings (from Prisma <code>onDelete: CASCADE</code>). Today's daily puzzle is protected by an explicit 409 check so the live leaderboard doesn't break.
Admin moderation	<code>username === 'admin'</code> can delete any puzzle.

13. R4 pass — Twilight Logic palette, permanent hero, premium feel

Area	Change
Twilight Logic palette	Canvas deepened to <code>#0a0a14</code> ; ink primary brightened to <code>#fafafe</code> . Four-colour accent system formalised: iris <code>#a78bfa</code> (brand), cyan <code>#22d3ee</code> (info/links/"You"), amber <code>#fbbf24</code> (Daily), rose <code>#fb7185</code> (destructive). All CSS vars exposed via <code>:root</code> .
Body multi-hue overlay	Three stacked radial gradients (purple, cyan, amber) at 4–6% opacity painted on <code>body</code> , <code>background-attachment: fixed</code> . Adds atmosphere without distraction.
Cage palette	12 riso-print-harmonious colours: coral, orange, amber, lime, emerald, cyan, sky, indigo, iris, fuchsia, pink, rose. Used at 10–12% opacity.
Hero grid — permanent + 6 layers	Idle float and slow tilt now run continuously via <code>useMotionValue</code> + <code>useTransform</code> . Mouse rotation composes on top instead of replacing idle. Six layers in 3D: ground shadow (translateZ -60 px), color glow (-22 px), glass backplate (-6 px), main cell grid, specular highlight, cursor-tracked light. Plus drifting particle layer (iris/cyan/amber dots). <code>prefers-reduced-motion</code> → static fallback.
Cells with depth	Filled cells translateZ 8 px with inset gradient + drop-shadow + cage-colour inner glow; empty cells sit flat with cage-colour inset glow.
Page transitions	New <code>src/app/template.tsx</code> re-mounts on route change and runs a 350 ms fade-and-rise animation. Reduced-motion users see no animation.
AnimatedSection helper	Reusable <code><AnimatedSection></code> for scroll-triggered fade-and-rise sections (used in <code>/play</code>).

Area	Change
Daily hero card	Wide card at top of <code>/play</code> with pulsing amber ambient glow, 180 px tilted 3D mini-grid, big "Play today" CTA.
3D mini-grid previews	Cards now show a 144 px MiniGridPreview rotated 8°/−8° with raised filled cells, cage-colour inset borders + outer specular highlight; hovers to reduced rotation + scale 1.04.
Depressible number pad	New <code>.numpad-btn</code> style with subtle gradient, 3D inset shadows, <code>:active</code> <code>translateY(2px)</code> ; when a digit is the currently-selected cell's value the button gets an iris ring + glow. Each button shows the remaining count for that digit.
Cage completion glow	Live <code>satisfiedCageIds</code> detection in the Grid component: as soon as the user's last cell of a cage falls into place with the correct sum and no duplicate, all that cage's cells flash a green success ring (one-shot, 900 ms) and the sum label scales 1.15× with a soft text-shadow. A subtle "complete" tone fires from the sound provider.
Selection animation	Selected cell scales 1.02× with a cyan outline + 2 px ring offset + 24 px halo; locked givens get the same outline but no scale.
Cell placement animation	<code>cell-pop</code> keyframe now includes a <code>rotateX</code> hint (perspective comes from the cell's own transform stack).
Empty state personality	New <code><DotGridIllustration></code> (3×3 dots, pulse animation) replaces bland "No data" panels on <code>/stats</code> empty, <code>/my-puzzles</code> empty, and the daily leaderboard. Copy is warmer + has a clear CTA.
Micro polish	Custom 8 px scrollbar (iris on hover), accent focus rings via <code>:focus-visible</code> , iris-tinted text selection, <code>cursor: cell</code> on grid cells, <code>scroll-behavior: smooth</code> , shimmer keyframe for skeleton loaders.

14. Repository layout

```

src/
  app/                Next.js App Router pages + API route handlers
    api/              auth, puzzles, results, ratings, daily, stats
      play/, create/, daily/, leaderboard/, stats/, rules/
    components/       shared, sudoku, builder
    lib/              solver · validator · hint · scoring · auth · db · utils
    store/            gameStore (Zustand)
  prisma/schema.prisma Prisma schema (MySQL provider)
  sudoku.sql          Raw MySQL DDL (equivalent to Prisma schema)
  seed.ts             Admin user + 9 puzzles + today's daily
  tests/              Vitest suites + JSON fixtures
  docs/               test-protocol-INITIAL.pdf, test-protocol-FINAL.pdf, DOCUMENTAT
  tools/md2pdf.mjs     Markdown-to-PDF helper (Chrome headless)
  setup.sh / setup.bat One-shot environment provisioning

```